

Consumer Choice

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The budget constraint

- A budget constraint is a constraint on how much money (income, wealth) an agent can spend on goods.
- Let $m \geq 0$ denote available income. $(x_1, x_2) \in \mathbb{R}_+^2$ denotes a consumption bundle.
- We assume that we can observe the prices of the goods (p_1, p_2)
- The **budget set** is then described by

$$\{(x_1, x_2) \in \mathbb{R}_+^2 \mid p_1 x_1 + p_2 x_2 \leq m\}$$

- Slope of the budget line $\frac{dx_2}{dx_1} = -\frac{p_1}{p_2}$

The consumers Utility Maximization Problem (UMP)

- We will model consumer choice **as if** he were maximizing utility.
- Do economists think that consumers actually maximize utility? **NO**.
- However, the model will yield some testable predictions that will actually conform to reality.
- Is this a good model? Depends on what you are analyzing.
- Advantage: wide range of applications under the same fundamental mathematical model.

Definition (Utility Maximization Problem)

Suppose that preferences over bundles $(x_1, x_2) \in \mathbb{R}_+^2$ are represented by $u(x_1, x_2)$, prices are (p_1, p_2) and income is m

$$\begin{array}{ll} \text{maximize} & u(x_1, x_2) \\ \text{subject to} & p_1 x_1 + p_2 x_2 \leq m \end{array} \quad (\text{UMP})$$

Well-behaved preferences

- To solve the UMP, we will assume some properties of the utility function that represents the consumer preferences.
- The purpose of the following assumptions is to make the analysis simpler.
- The properties we will assume are:
 - $u : \mathbb{R}_+^2 \rightarrow \mathbb{R}$ is differentiable, so that the partials exist and we can calculate the $MRS = u_1/u_2$.
 - Monotonicity: "*the more the better*". Implies that marginal utilities are always positive.
 - Convexity of preferences or a tastes for balanced consumption: equivalent to a decreasing MRS. Strict convexity implies a strictly decreasing MRS.
- We will say that a preferences that satisfy the three properties are **well-behaved**.

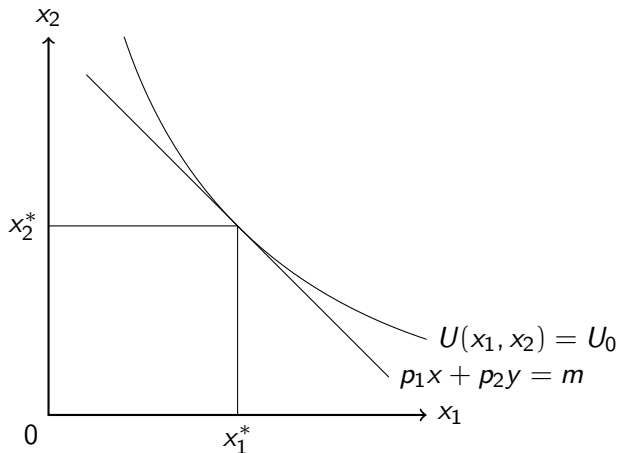
Graphical Analysis: Intuition

At an **interior** optimum, the slopes of the utility function and the budget line need to be equal. Thus

$$-\frac{p_1}{p_2} = -\frac{u_1}{u_2}$$

$$\frac{p_1}{p_2} = \frac{u_1}{u_2}$$

$$\frac{p_1}{p_2} = MRS$$



A detour on Unconstrained Optimization

Let f be a real-valued function defined over $\mathbb{R}^n$, that means $f : \mathbb{R}^n \rightarrow \mathbb{R}$.

Definition

The function f has a **local maximum** at x^* if there exists a neighborhood I such that it contains x^* and $f(x^*) \geq f(x)$ for all $x \in I$.

Definition

The function f has a **global maximum** at x^* if $f(x^*) \geq f(x)$ for all x in the domain of f .

First and Second order conditions: Single variable case

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a real-valued single variable function.

Theorem (First Order Necessary Condition for maximum/minimum)

Let f be a real-valued function of one variable. Suppose that f has a local maximum/minimum at some point x^ and suppose that $f'(x)$ exists. Then $f'(x^*) = 0$.*

Theorem (Second Order Sufficient condition for a maximum(minimum))

Let f be a real-valued function of one variable. Suppose that $f''(x^)$ exists and suppose that $f'(x^*) = 0$ and $f''(x^*) < 0$ ($f''(x^*) > 0$). Then f has a local maximum(minimum) at x^* .*

Multivariate function Unconstrained maximization

Theorem (First Order Necessary Condition for maximum/minimum)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Suppose that f has a local maximum/minimum at some point x^* and suppose that f is differentiable. Then $\frac{\partial f(x_1, \dots, x_n)}{\partial x_i} = 0$ for all $i = 1, \dots, n$

- To get second order sufficient conditions for maxima (minima), we need to be able to talk about when a multivariate function f is concave (convex) $\rightarrow$ left for a more advanced course.
- Most problems we face in the course will satisfy the second order sufficient conditions.

Back to constrained Optimization

Definition (First order conditions for interior maximum)

Suppose preferences are represented by the utility function $u(x_1, x_2)$. If (x_1^*, x_2^*) is an interior constrained maximum, then we must have

$$\frac{p_1}{p_2} = MRS(x_1^*, x_2^*)$$

Definition (Second order conditions for interior maximum)

A decreasing MRS is sufficient to guarantee that the point identified by the FOC $MRS = p_1/p_2$ is indeed the constrained maximum. Moreover, a strictly decreasing MRS yields a **unique maximizer**. What conditions on the utility function guarantee a decreasing MRS? If preferences are convex we have seen that they can be represented with a utility function which will have a decreasing MRS.

Optimization by Substitution

- Suppose that the consumer has **monotonic preferences**, i.e. the more the better.
- Then we know for sure that he will try to exhaust his income.
- Thus, the restriction of the budget constraint will hold with equality:
 $p_1x_1 + p_2x_2 = m$.
- Thus we can substitute $x_2 = \frac{m}{p_2} - \frac{p_1}{p_2}x_1$ into the utility function to get

$$u(x_1, x_2) = u\left(x_1, \frac{m}{p_2} - \frac{p_1}{p_2}x_1\right)$$

- We can treat this problem as a single variable unconstrained maximization problem, will get that the FOC is indeed $MRS = p_1/p_2$.
- By calculating directly the second order derivative of the univariate function $u\left(x_1, \frac{m}{p_2} - \frac{p_1}{p_2}x_1\right)$ and evaluating at the optimum, we can get the conditions on the second order derivatives of u directly which are equivalent to u having a decreasing MRS . *Note: Won't pursue this any further. If curious, check the advanced reference Nicholson and Snyder or ask me.*

The Lagrangian Method for constrained optimization

- There is a second way (and the preferred one for the course) to solve the UMP and in general optimization problems with equality constraints.
- Recall that due to monotonicity, we can state the UMP as:

$$\begin{array}{ll}\text{maximize} & u(x_1, x_2) \\ \text{subject to} & p_1x_1 + p_2x_2 = m\end{array}$$

- The method we will use uses **Lagrange Multipliers**
- We start by defining an auxiliary function, the **Lagrangian**:

$$L = u(x_1, x_2) - \lambda(p_1x_1 + p_2x_2 - m)$$

- It introduces a new variable λ , called the **Lagrange multiplier** which multiplies the constraint.

The Lagrangian Method to solve the UMP.

$$L = u(x_1, x_2) - \lambda(p_1x_1 + p_2x_2 - m)$$

- Lagrange's theorem says that an optimal choice (x_1^*, x_2^*) must satisfy the first-order conditions:

$$\frac{\partial L}{\partial x_1} = u_1(x_1^*, x_2^*) - \lambda p_1 = 0$$

$$\frac{\partial L}{\partial x_2} = u_2(x_1^*, x_2^*) - \lambda p_2 = 0$$

$$\frac{\partial L}{\partial \lambda} = p_1x_1 + p_2x_2 - m = 0$$

- Using the first two conditions with respect to x_1 and x_2 and using them to eliminate λ yields

$$MRS = \frac{u_1(x_1^*, x_2^*)}{u_2(x_1^*, x_2^*)} = \frac{p_1}{p_2}$$

which is the same condition we obtained before!

The Lagrangian Method: remarks

- Notice that the derivative of L with respect to λ is simply the equation of the budget line. Remember that monotonicity guarantees that the consumer will end up choosing a bundle that is in the budget line.
- Recipe: form the Lagrangian, take first order derivatives with respect to the original variables x_1, x_2 and the new variable λ to get a system of **three equations in three unknowns**.
- The system of equations is solved in terms of the other parameters: prices p_1, p_2 and income m .
- The optimal solutions: $x_1^*(p_1, p_2, m)$ and $x_2^*(p_1, p_2, m)$ are called **demand functions**.
- The optimal solution to the multiplier $\lambda^*(p_1, p_2, m)$ holds a special meaning, which we will cover more in depth later.

Class examples: UMP for different utility functions

- Cobb Douglas utility: $u(x_1, x_2) = x_1^\alpha x_2^\beta$. *Using the natural log as a monotonic transformation, will yield a simpler problem.*
- (time permitting) CES utility: $u(x_1, x_2) = (\alpha x_1^\delta + \beta x_2^\delta)^{\frac{1}{\delta}}$.

What can go wrong? Corner solutions, non differentiability, Bliss points

- The condition $MRS = p_1/p_2$ helps identify an interior optimum.
- What happens when it is optimal to consume positive quantities of one good, but none of the other?
 - Perfect substitutes: $u(x_1, x_2) = \alpha x_1 + \beta x_2$. *Better to use a graphical method.*
 - Quasilinear utility: $u(x_1, x_2) = v(x_1) + x_2$, for example, take $v(x_1) = \ln(x_1)$. *Here we can use the Lagrangian method, but we must watch out for a **corner solution**, since recall that only positive consumption bundles are allowed.*
- What happens if preferences are non-differentiable? Perfect complements $u(x_1, x_2) = \min(\alpha x_1, \beta x_2)$
- On the other hand, what if the preference do not satisfy the more the better? *If preference have a **bliss point** it is possible that the optimum can happen **inside the budget set**. If the bliss point is affordable, that is the optimum. If it is not affordable, use Lagrangian.*